

Auditing Small-Model Adaptation for Product Search and Catalog Understanding

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Abstract

Small language models offer a practical route to product-search and catalog decisions, but apparent task gains can depend on data exposure, scoring, and checkpoint selection. We study a 1.7-billion-parameter model adapted for relevance, product identity, functional relations, compatibility, and attribute extraction. An executable retrospective audit reconstructs the available data versions, training trajectories, comparator inputs, and serving traces. It finds that 25 of 28 functional test rows reuse training scenarios, that a substring scorer accepts opposite compatibility labels, and that nine checkpoints were evaluated on the same named test set. A dedicated extraction adapter obtains brand/color F1 of 1.000/0.955 on 200 listings under the historical scorer, but two test texts overlap training and missing paired predictions prevent a definitive significance analysis. A relevance alternative attains 0.5115 accuracy yet only 0.436 macro-F1. We formalize a comparison contract and provide a strict evaluation implementation. New GPU development experiments retain complete outputs, pair the base and existing adapters on identical examples under two interfaces, and add a text-extraction comparator and bounded HTTP serving measurements. Final tests remain unscored. The results establish a narrow extraction signal and concrete limits on multitask and efficiency claims, rather than universal superiority over general-purpose models.

CCS Concepts

• Information systems → Information retrieval.

Keywords

product search, entity matching, small language models, evaluation, data leakage, reproducibility

1 Introduction

Product search depends on several kinds of understanding. A search engine must determine whether an offer satisfies a query; a catalog must determine whether two offers refer to the same purchasable variant; and recommendation or filtering systems must distinguish functional relationships from technical compatibility. Attribute extraction supplies structured evidence for each of these decisions. A compact model covering these tasks could reduce the engineering burden for retailers with limited compute.

Their common vocabulary does not make their labels interchangeable. Two different headphones can both satisfy a

query. A charger can complement a laptop without supporting its voltage or connector. A plausible color can be absent from the listing. These distinctions make broad commerce adaptation a demanding evaluation problem: even a numerically high score can fail to establish the operational behavior its label suggests.

This paper examines a practical sequence of shared and specialist QLoRA experiments, then subjects the preserved artifacts to an executable retrospective audit. The study is deliberately explicit about this chronology. The original runs were not a prospectively controlled factorial experiment; a later audit cannot recover missing predictions or make a repeatedly inspected test set independent. It can identify the conclusions that remain supported and provide concrete controls for the next experiment.

We ask three questions. **RQ1:** What task behavior do the saved adaptation results establish? **RQ2:** Which comparisons remain interpretable after checking data exposure, evidence equivalence, scoring, and selection? **RQ3:** What do the recorded training and serving traces establish about efficiency? These questions concern information retrieval because product-understanding labels become inputs to ranking, catalog consolidation, and filtering. We do not equate pair classification with measured improvement in an end-to-end search system.

The contribution has three parts. First, we reconstruct the five task formulations and their shared/specialist training record, exposing minority-task and versioning effects hidden by aggregate scores. Second, we introduce an executable comparison audit that checks scenario and text overlap, scorer counterexamples, checkpoint exposure, and trace completeness. Third, we quantify what the surviving extraction aggregates identify, including the range of paired correctness tests compatible with those aggregates. We provide strict parsing and grouped evaluation code for subsequent runs. These are empirical and reproducibility contributions; we do not claim to invent low-rank adaptation, commerce instruction tuning, or a new foundation-model architecture.

2 Related Work

Commerce instruction tuning. eCeLLM [1] studies commerce instruction tuning with held-out task and category evaluations and generalist/specialist comparisons. EcomGPT [2] constructs instructions through task transformations and chain-of-task tasks. These works establish that combining commerce tasks and synthetic instructions is prior art. Our local eCeLLM measurements do not reproduce its published suite, and our results cannot establish superiority over its generalization claims. CASLIE [3] adds

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context-sensitive captions and caption-quality selection; its multimodal evidence differs from our text-only setting.

Compact systems and task-specific alternatives. Licardo and Tanković [4] study small-model commerce adaptation and deployment optimization. Their measured hardware and quantization comparisons highlight why parameter count alone cannot establish serving efficiency. Ditto [7] is a relevant supervised entity-matching baseline. GLiNER2 [8] illustrates compact, schema-driven extraction and classification. Comparisons should include trained task-specific systems as well as general-purpose APIs; an off-the-shelf encoder result is not an upper bound on its architecture.

Data and methodology. ESCI [5], WDC Products [6], and ABO [9] provide the source data. Their publisher-defined tasks and splits differ from our local subsets. LoRA [10] and QLoRA [11] supply the adaptation mechanism. GradNorm [12] and PCGrad [13] provide established alternatives for balancing multitask learning; neither was evaluated here. Leakage analysis [14] and behavioral testing [15] motivate our audit. Applying these principles to this experimental record is a concrete contribution, not a claim that leakage prevention or scorer testing is itself new.

3 Tasks and Data Provenance

3.1 Task semantics and artifact scope

Relevance predicts *exact*, *substitute*, *complement*, *irrelevant* for a query and listing. Identity predicts *same*, *distinct* for two listings at the purchasable-variant level. Functional relation predicts *substitute*, *complement*, *unrelated*. Compatibility predicts *compatible*, *incompatible*. Extraction returns JSON with *brand* and *color*. Although project schemas allow an unknown relation, the recorded experiments do not train or evaluate calibrated abstention.

A shared four-task adapter, an extraction specialist, and a later functional specialist are distinct parameter states. A letter-label relevance adapter is another experimental state. Consequently, system-level task coverage does not establish that a single parameter state is strong on all tasks. This distinction is central to interpreting a horizontal model claim.

The audit covers original data, source, and report files together with archived trainer states, adapter configurations, and checkpoint histories. Generated tables are derived from these inputs and checked against a source-hash manifest. Historical files are preserved rather than rewritten to agree with the manuscript. Missing raw predictions and environment records remain missing evidence.

3.2 Mixtures and independent observations

The current shared mixture contains 18605 rows: 14084 train and 4521 development. Table 1 distinguishes this state from the saved first-run development file. Functional and compatibility development originally contain only one class each. Their perfect raw accuracy therefore does not establish multiclass performance.

Table 1: Current mixture versus saved first-run development data. The final column is the number of represented labels in the saved split.

Task	Train	Dev	Old dev	Labels
Relevance	8000	1000	1000	4
Identity	6000	3500	3500	2
Functional relation	52	13	15	1
Compatibility	32	8	4	1

The scaled shared run uses 46,086 training and 7,519 development rows, including 40,000/4,000 relevance rows and 6,000/3,500 identity rows. It preserves earlier minority-task assignments of 50/15 functional and 36/4 compatibility rows. Current files therefore cannot stand in for every historical run. Relevance train/dev query identifiers and normalized full inputs are disjoint in the audited mixtures, but this does not prove product-family, merchant, or pretraining independence. The shared identity development set contains 3,000 distinct and 500 same labels; an always-distinct classifier achieves 0.857 accuracy.

API comparison samples contain 60 relevance rows, balanced at 15 per class, and 50 identity rows, balanced at 25 per class. Their formatted inputs do not exactly match the current shared train/dev inputs. This establishes a sample mismatch, not a paired comparison. Subtracting an API score on these rows from an adapter score on thousands of different rows is not a model effect estimate.

3.3 Synthetic scenario dependence

The synthetic process specifies a scenario and intended relation, generates descriptions with Claude Haiku, and checks them with GPT-5-mini. The judge receives the intended label and scenario. This is a cross-model consistency check rather than blinded human adjudication. Neither naturalness nor factual validity was measured by human raters.

The initial functional and compatibility datasets contain 65 and 40 rows. An expansion audits 124 functional generations and retains 122, producing 187 rows over 63 scenario objects. The final split assigns 131/28/28 rows to train/dev/test. Although normalized generated text is disjoint, 26/28 dev and 25/28 test rows share a scenario object with training. The split separates paraphrases more often than underlying decisions.

An earlier split repair introduces further exposure: 11/13 corrected functional dev texts and 6/8 corrected compatibility dev texts appear in the earlier training assignment preserved in the scaled mixture. Relabeling their partition cannot erase prior model access. These sets diagnose label coverage but cannot establish generalization to unseen scenarios.

3.4 Extraction grounding and overlap

ABO-derived extraction has 9,396 train, 300 dev, and 200 evaluation rows. Training contains 9,359 unique item identifiers and 9,319 unique texts. Train/dev overlap comprises four identifiers and two exact texts; train/evaluation has no identifier overlap but two exact texts. An identifier-only exclusion rule misses this failure.

Every brand value occurs literally in its supplied text under case-insensitive matching. Color values do so for 8,390/9,396 train, 256/300 dev, and 184/200 evaluation rows. A nonliteral value may reflect metadata normalization rather than an incorrect label. Nonetheless, the benchmark mixes span extraction with interpretation. An exact-grounding claim requires a separate policy for extraction, normalization, and inference, with evidence sufficiency checked under that policy.

4 Adaptation and Evaluation Methods

4.1 Training implementation

The adapters use the upstream Qwen3-1.7B identifier. Base weights are quantized with NF4 and double quantization, with bfloat16 computation. For a projection,

$$W' = W_{\text{base}} + \frac{\alpha}{r}BA, \quad (1)$$

where A and B are trainable rank- r matrices. Attention query, key, value, and output projections and feed-forward gate, up, and down projections are adapted. Rank 16, scale 32, and dropout 0.05 are preserved for the initial and specialist adapters; the scaled shared configuration uses rank 32 and scale 64.

Instruction and input prefix positions and padding are masked in the causal loss. Only target completion positions contribute:

$$\mathcal{L} = -\frac{1}{\sum_i |y_i|} \sum_i \sum_{j=1}^{|y_i|} \log p_{\theta}(y_{ij} \mid x_i, y_{i,<j}). \quad (2)$$

Concatenated examples are truncated from the right. The historical artifacts do not report how often this removes all supervised target tokens. The registry contains an intended base revision, but executed loader calls omit a revision argument and archived adapter configurations record a null revision. The intended pin therefore does not certify every historical download.

The weighted functional and relevance variants assign inverse-frequency weights $N/(Kn_c)$ to selected vocabulary tokens. This differs from example-level class weighting when a label spans multiple tokens. The letter-label variant asserts that each of its four label codes is one token; the functional variant weights the first token of verbal labels. These details affect any interpretation of a weighting ablation.

All configurations use a cosine schedule; specialist scripts record warmup and extraction uses gradient checkpointing. Table 2 reports optimizer steps, which are not epochs. Preserved states place shared v1 at epoch 0.454, extraction at 2.041, functional at 66.71, and relevance letters at 0.48.

Table 2: Preserved training configurations. Effective batch is 16 on one device. Shared scripts changed in place; this is a reconstruction, not a frozen historical environment.

Run	Rank	Steps	Length	LR
Shared v1	16	400	256	10^{-4}
Shared v2	16	800	256	10^{-4}
Shared v3	32	1,200	256	10^{-4}
Relevance letters	16	1,200	256	$3 \cdot 10^{-5}$
Functional	16	600	320	$2 \cdot 10^{-5}$
Extraction	16	1,200	384	10^{-4}

Dataset repetition and different row counts make a common step count an inadequate compute control.

4.2 A comparison contract

We represent a result’s evaluation contract as

$$\mathcal{C} = (D, I, X, Y, P, G, S, H), \quad (3)$$

where D identifies the source/split version, I the row identifiers, X the supplied evidence, Y the gold ontology, P the prompt, G the decoding budget, S the scorer, and H the selection history. Model configuration and execution environment are recorded separately. A paired quality comparison requires equal task, rows, evidence, and scoring, with declared prompt/decoding differences. The history component records whether evaluation outcomes influenced prompts, thresholds, checkpoints, or retraining.

This contract makes common failure modes explicit. Same task names do not imply the same sample; the same source corpus does not imply the same gold interpretation; and a file named test is not an independent holdout after selection. It is an engineering representation of established experimental requirements rather than a statistical guarantee or a new learning objective.

4.3 Scoring and statistical interpretation

The original shared scorer tests whether the gold string appears in the prediction. Thus `compatible` is accepted within `incompatible`; `same` is accepted within `not same`. Extraction uses either-direction string containment, so a partial color can match a longer value. We report historical scores under those rules and do not silently recast them as strict exact match.

The new evaluator requires complete labels after conservative normalization, rejects duplicate or unexpected JSON keys, distinguishes null from a missing response, and reports exact and normalized field matches separately. Every expected ID must have one output record; failures remain in the denominator. Human-readable explanations or ambiguous multi-label outputs are invalid under a label-only contract.

For future paired measurements, keys for shared queries, entities, or families are joined into connected components

Table 3: Letter-label relevance performance on its 4,000-row local evaluation. P/R/F1 are class metrics, not ranking metrics.

Label	Gold	Pred.	P	R	F1
exact	1753	1608	0.616	0.565	0.589
substitute	1381	1939	0.418	0.587	0.489
complement	175	86	0.407	0.200	0.268
irrelevant	691	367	0.572	0.304	0.397

before resampling. The implementation reports a cluster-bootstrap interval for binary or joint correctness only when at least 20 components are available. A row-iid McNemar calculation is explicitly diagnostic; it does not replace group-aware uncertainty or establish a field-F1 effect. These controls also support the exploratory paired development measurements reported below.

5 Results and Audit Findings

5.1 Shared adaptation and relevance

Recomputed logged first-run accuracy is 0.527 on 1,000 relevance rows and 0.916 on 3,500 identity rows. Subsequent shared runs report 0.491/0.914 and 0.50275/0.887714 for relevance/identity; the last relevance sample has 4,000 rows. Mixture size, rank, sampling, steps, and evaluation populations change together. The trajectories do not isolate any one cause and cannot prove catastrophic forgetting or reject multiple causal hypotheses.

The separate letter-label relevance run yields 0.5115 accuracy on 4,000 rows, but 0.436 macro-F1 and 0.414 balanced accuracy. Table 3 shows the minority-class problem: complement recall is 0.200, and substitutes are predicted 1,939 times against 1,381 gold instances. A single aggregate accuracy masks this behavior.

On the separate 60/50-row API samples, recorded relevance/identity accuracies are 0.4667/0.780 for Haiku, 0.5833/0.920 for Sonnet, 0.4833/0.800 for GPT-5-mini, and 0.4667/0.840 for GPT-5. These are historical report identifiers, not verified current endpoint snapshots. The artifacts lack per-response model identifiers and complete raw response journals. Output caps also differ: the Anthropic call sets ten tokens while the OpenAI call does not impose the same cap. No corresponding shared-adaptor predictions survive on those exact rows. These numbers provide context, not a frontier-model ranking.

5.2 Functional specialization and test exposure

The published functional checkpoint reaches 23/28 accuracy (0.821) and 0.780 macro-F1. It correctly classifies 14/14 complements, 7/10 substitutes, and 2/4 unrelated pairs. All five errors become complement predictions. The small, dependent test population cannot establish broad category or scenario transfer.

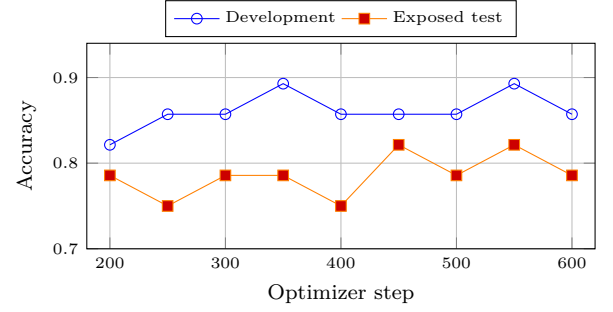


Figure 1: All available functional checkpoint comparisons. Repeated test access is part of the finding; this curve is not a selection protocol to emulate.

Nine checkpoints, from step 200 through 600, were evaluated on the same named test set (Figure 1). The earliest development maximum occurs at step 350 and obtains 22/28 test accuracy; the published step-450 checkpoint obtains 23/28. Because test outcomes were available during checkpoint comparison, the published result is an exposed-test diagnostic. A retrospective alternative selection rule cannot restore an unseen test set.

The functional API comparison uses a different 13-row set and reports 11/13 accuracy. Its parser takes the first two newline-delimited fields, leaving the second listing empty in two rows. The local eCeLLM run preserves 28 generations: ten are **a:** **b**, and eighteen are malformed JSON fragments. None is a valid single-label response under the required ontology. A first-character parser manufactures labels from them. This demonstrates an interface failure in that run; it is not evidence that the underlying model lacks functional knowledge.

5.3 Extraction: a signal with bounded interpretation

Table 4 reports the shared 200-row extraction sample. Under the historical permissive metric, the dedicated adaptor obtains brand F1 1.000 and color F1 0.955. It ties both GPT systems on brand. The strongest API color result is Haiku at 368/391 $\approx$ 0.9412, so the observed difference is 1.38 percentage points. Comparing only with GPT-5 would overstate the best-baseline margin.

Matched inputs are necessary but insufficient: the saved reports contain aggregates rather than aligned per-row predictions. The two training-overlap texts cannot be removed and rescored reliably, and paired field-F1 confidence intervals cannot be reconstructed. The result supports further investigation of specialization, not a definitive clean-data win.

We can nonetheless characterize what the binary correctness marginals imply. Let x be the number of listings on which both the adaptor and Haiku are correct on color. With 191 and 184 correct among 200,

$$175 = 191 + 184 - 200 \leq x \leq 184. \quad (4)$$

Table 4: Historical extraction results on 200 listings, under the permissive field scorer. Color counts are TP/FP/FN. Two evaluation texts occur in training. These are not decontaminated strict-match results.

System	Brand F1	Color F1	Color counts
Dictionary	0.880	0.754	133/20/67
GLiNER2	0.777	0.854	167/24/33
GLiNER2.5	0.753	0.871	166/15/34
Haiku	0.965	0.941	184/7/16
Sonnet	0.997	0.925	180/9/20
GPT-5-mini	1.000	0.926	182/11/18
GPT-5	1.000	0.931	183/10/17
Dedicated adapter	1.000	0.955	191/9/9

The discordant counts are $b = 191 - x$ and $c = 184 - x$. For each feasible x , the two-sided exact McNemar value is

$$p(x) = \min \left(1, 2 \sum_{j=0}^{\min(b,c)} \binom{b+c}{j} 2^{-(b+c)} \right). \quad (5)$$

Across feasible pairings it ranges from 0.015625 to approximately 0.229523. Thus identical marginal totals permit either side of the conventional 0.05 threshold. This is a partial-identification calculation for binary correctness under row independence, not an observed paired test, a bootstrap interval, or an F1 significance result. Missing pairing materially limits the claim even before contamination is considered.

5.4 Coverage of relevant comparators

A broader local comparator sweep is incomplete. Some embedding/relevance results collapse four labels into binary relevance and tune thresholds on the evaluated sample. Those scores cannot be inserted into a four-class table. A trained Ditto-style matcher, appropriately tuned compact extraction systems, and a completed NuExtract comparison remain missing. The same is true of public-protocol results for the full ESCI and WDC benchmarks. We treat these as evidence gaps rather than explain them away through architecture labels.

6 Training and Serving Evidence

6.1 What the logs measure

The preserved loss traces show optimization progress but do not establish generalization. Functional and letter-label states record approximately 389.2 and 1,004 seconds of trainer runtime, respectively. These exclude missing campaign runs, data construction, evaluation, provisioning, idle time, and unsuccessful trials. They are insufficient for a total training-cost comparison. The available configurations also do not constitute fixed-token or fixed-dollar ablations.

The service uses three separately loaded base-model instances for shared matching, extraction, and functional specialization. Locks protect lazy initialization, but do not establish concurrent scheduling efficiency, adapter multiplexing, or GPU memory sharing. A health route reports a process response without demonstrating that every model is loaded and ready for inference.

The available before-change load CSV contains one health completion and zero inference completions. The after-change CSV contains no completed requests. Neither provides an inference latency distribution. Pending, timed-out, or missing requests cannot be treated as zero-latency successes. A rate of 20 tokens per second and a price of \$0.74 per hour appear as assumptions, not measured hardware or billing results. These historical traces establish no throughput, p95 latency, or serving-cost advantage. The separate follow-up below measures a bounded GPU HTTP prototype.

6.2 Quality-constrained serving measurement

A useful serving objective should count successful task outcomes rather than generated tokens alone. For a predeclared latency bound L and observation window of duration T , define

$$\text{QPS}_L = \frac{\sum_i \mathbb{I}[\text{correct}_i \wedge \text{valid}_i \wedge \ell_i \leq L]}{T}. \quad (6)$$

Cost per such completion is the billed cost over that same window divided by the numerator; it is undefined if there are no qualifying completions. This definition makes schema failures, wrong predictions, cold starts, and queueing relevant to the measured service. It is a prospective measurement criterion, not a new serving algorithm or a quantity estimated by the historical traces.

A valid comparison must fix workload IDs, input/output lengths, task mix, concurrency or arrival process, precision, software versions, and warm/cold conditions. It should retain every request’s start/end/status, separate warmup, drain outstanding requests to a deadline, and report censored work. Hardware cost requires actual instance specifications and billed time. None of these measurements can be inferred solely from a 1.7B parameter count or a nominal GPU listing.

7 Prospective Protocol and Development Pilot

7.1 A new evaluation population

A separate real-source suite contains 17,997 training, 1,495 development, and 3,000 test rows. Relevance uses whole ESCI queries with cross-partition product exclusions; identity uses WDC entity groups; extraction uses ABO item, text, image, and manufacturer-identifier proxies. Previously exposed examples are excluded from new evaluation using available identifiers and group keys. The manifest records zero cross-partition group-key overlap. This is a local subset protocol, not the publishers’ complete benchmark or a guarantee against pretraining contamination.

Table 5: Untuned base on the new development suite. These strict metrics use a different population from historical adapter results. Norm. denotes Unicode, case, and whitespace normalization.

Task	Rows	Metric	Value
Relevance	595	Accuracy	0.3361
		Macro-F1	0.2540
Identity	500	Accuracy	0.3340
		Macro-F1	0.2865
Extraction	400	Brand F1 (exact)	0.7250
		Color F1 (exact)	0.7181
		Brand F1 (norm.)	0.7275
		Color F1 (norm.)	0.7914
		Joint acc. (norm.)	0.5800

Development contains 595 relevance, 500 identity, and 400 extraction examples. Relevance comprises 278 exact, 225 substitute, 19 complement, and 73 irrelevant rows; identity has 148 same and 352 distinct rows. Majority-class development accuracy is therefore 0.4672 and 0.7040, respectively. These are descriptive controls, not models selected using test outcomes. Extraction requires literal, nonempty brand and color values and cannot test missing-field abstention.

Machine checks do not approve training. Independent human evidence-sufficiency and naturalness review remains outstanding; no new training is performed. New synthetic relation and compatibility tasks remain blocked pending independently reviewed, scenario-disjoint data. Test labels are separated from inference inputs; structural verification is allowed, but no test predictions or test-based selection are made.

7.2 Pinned-base GPU baseline

We evaluate unadapted Qwen3-1.7B revision 70d244cc... on all development examples (Table 5). Full revision, model-file hashes, prompt/scorer hashes, raw outputs, and software versions accompany the artifact. We use one RTX 3090 with 24 GiB device memory, PyTorch 2.8.0 with CUDA 12.8, Transformers 4.57.6, bfloat16, and batch size one. Greedy generation disables thinking and permits 64 new tokens. Inputs exceeding 2,048 tokens are rejected rather than truncated; failures remain in denominators. The seed is 20260928.

All 1,495 outputs satisfy the required schemas, but schema compliance does not imply task competence. The model predicts same for 481/500 identity pairs and never predicts substitute on the relevance subset. Its classification accuracies fall below the descriptive majority controls. Extraction normalized joint accuracy is 0.5800. These behaviors motivate matched adapter evaluation. The separate follow-up reruns the base and existing adapters on the same examples under two explicitly distinguished interfaces.

The sequential evaluation loop takes 218.90 seconds, excluding loading, provisioning, and preparation. Per-row timings and generations are retained, but the run has no network

Table 6: Matched development primary scores: macro-F1 for relevance and identity; normalized exact joint accuracy for extraction. Identical rows within each task.

Task	Interface	Base	Adapter
Relevance	legacy	0.0000	0.3333
Relevance	chat	0.2568	0.2665
Identity	legacy	0.0000	0.8617
Identity	chat	0.3110	0.8886
Extraction	legacy	0.0000	0.9150
Extraction	chat	0.5825	0.9025

queue or concurrency experiment. It supplies neither production tail latency nor a serving-cost advantage. A preliminary nine-row execution check is excluded from this quality table. Neither run accesses the final test for scoring, and no paid frontier comparison is added.

8 Matched Development and HTTP Follow-up

The follow-up evaluates the pinned base and existing adapters on the same 1,495 development rows under two interfaces: original training completion prompts (legacy), and the common chat task specification. All complete paired runs use bfloat16, batch size eight, greedy decoding, a 64-token output cap and the strict scorer. Neither training nor final-test scoring occurs. An initial batch-one run was interrupted for runtime reasons; its partial journal is retained but excluded. Batching changes are shared by all paired candidates.

Interface and adaptation must be interpreted jointly (Table 6). Strict schema failures remain in the denominator; per-class and validity results accompany the raw outputs. The recorded historical training-source audit finds no exact-text matches on these development rows, but identity entity exposure remains unresolved. Thus this is a retrospective development comparison, not certified independent generalization. Paired binary/joint correctness diagnostics use connected groups and 2,000 exploratory bootstrap resamples; their intervals do not describe macro-F1 effects. No repeated-seed training effect or matched frontier-API ranking is established.

The unadapted base fails the strict output contract on every legacy-prompt row. Its zero scores measure interface failure, not absence of commerce knowledge; the chat comparison is essential to interpretation. Under chat, relevance macro-F1 changes only from 0.2568 to 0.2665, whereas identity changes from 0.3110 to 0.8886. The adapter predicts no complement labels in either interface. Extraction joint accuracy changes from 0.5825 to 0.9025 under chat. The connected grouping yields 33 relevance, 5 identity and 367 extraction components. We suppress the identity bootstrap interval because fewer than 20 components do not support that uncertainty estimate. BF16 greedy outputs can depend

Table 7: GPU loopback HTTP prototype. B: maximum batch size; C: client concurrency. QPS counts completed requests; correctness is over all attempts. p95 is the mean of three within-run percentiles, not a pooled percentile.

B	C	QPS	Correct	p95 (s)
1	1	9.01	0.900	0.116
1	4	8.96	0.900	0.466
1	8	9.06	0.900	0.933
8	1	8.57	0.900	0.124
8	4	29.11	0.905	0.148
8	8	40.56	0.900	0.287

on batching; the batch-eight base is rerun rather than substituted by the earlier batch-one pilot. These findings distinguish task adaptation, label compliance and interface sensitivity rather than supporting a universal gain.

The NuExtract-2.0-2B comparison uses the native schema interface for text, with verbatim-string brand/color fields and no demonstrations or prompt search. On the same 400 extraction rows it obtains normalized joint accuracy 0.4675, normalized brand/color F1 0.5667/0.8283, and invalid rate 0.0000. The revision, prompt and raw outputs are retained. This closes one missing comparator run, not broader baseline coverage.

The shared identity adapter is served through a bounded HTTP queue on one RTX 3090, comparing batch limits one and eight with at most 5ms batching wait. Each cell has three repetitions of 200 requests on the same development trace; 50 warmup requests per engine/repetition are excluded. Generation permits eight tokens. Client and server share the pod, and the load is closed-loop, so these results do not identify open-loop overload behavior or external network latency. The smaller workload is an explicit budgeted departure from the prospective larger serving study; tail estimates are exploratory.

Quality and throughput are reported together (Table 7); all attempted responses, errors and timestamps are retained. Window-specific cost estimates use the quoted GPU rate of \$0.22/hour and exclude provisioning, idle time, storage and network. They are not invoices, production SLO costs or an API cost advantage. The reference queue is not a new serving algorithm or a comparison with an established optimized inference engine.

9 Implications, Limitations, and Reproducibility

Controls implemented for the next experiment. The new isolated evaluation code validates exact ontologies and one-to-one output joins, detects cross-partition group overlap, preserves errors in denominators, and generates paired correctness diagnostics from connected groups. The new data builder excludes historical exposure using available query, entity, text, and product-family proxies. Machine checks fail

closed when there are insufficient independent rows or missing label support. Test labels are stored separately from inference inputs. File separation is an operational control, not a claim that the experimenter cannot access them.

Product-family grouping based on manufacturer identifiers and shared images is only a proxy. It requires inspection for generic identifiers and does not prove that all semantically related variants have been found. Similarly, public-corpus provenance does not guarantee realistic coverage for every retailer. The current extraction policy emphasizes literal, nonempty fields and does not establish performance on missing attributes, conflicting descriptions, multilingual catalogs, or merchant-specific taxonomy changes.

Experiments needed for stronger claims. The development follow-up addresses matched base/adaptor interfaces and one extraction comparator. Decisive final evaluation still requires task-appropriate compact baselines, commerce models and API systems on the same frozen inputs. Development data may control prompts, loss weights, and stopping; test outcomes may not control further training in the same reported experiment. Multiple seeds and fixed-data/token budgets are needed to attribute gains to a mechanism. In particular, scenario diversity versus paraphrase volume should be varied independently, with blinded human checks of label support and naturalness. Group-disjoint held-out scenarios and categories are necessary to test broad commerce transfer.

Beyond the bounded HTTP prototype reported here, serving requires open-loop GPU measurements, broader task coverage and actual invoiced cost. A single-GPU sequential development run cannot establish an optimal RunPod instance or production throughput. Adaptive training and retention safeguards remain research hypotheses until a controller is implemented and compared with static and established multitask controls. We do not infer universal retailer suitability, absence of catastrophic forgetting, or a guaranteed win against future models.

Scope of reproducibility. The retrospective audit regenerates tables and curves from saved evidence, exercises actual scorer counterexamples, and checks source hashes. It does not recreate training environments that were never archived. For new runs, immutable model revisions, tokenizer/config hashes, adapter hashes, prompt/scorer versions, raw outputs, seeds, and dependency/device records must accompany each metric. Public source datasets retain their licenses and attribution; local subsets must not be mislabeled as official leaderboard submissions. A submission artifact must be distributed through an anonymous channel; an author-identifying public repository is not an anonymous review artifact.

10 Conclusion

This study shows how a seemingly broad commerce-model result narrows when its evaluation history is reconstructed. Relevance errors concentrate in minority classes; functional

evaluation reuses scenarios and exposes the test during selection; and a promising extraction margin cannot be converted into a clean paired significance claim from the surviving aggregates. Historical serving traces provide no completed inference distribution. New development measurements distinguish prompt-interface and adapter behavior, and a bounded HTTP prototype provides a separate quality-aware serving measurement. The executable audit and evaluation implementation make these limits inspectable and identify the measurements required for stronger product-search and catalog-understanding claims.

11 Ethical Considerations

Incorrect identity merges can conflate incompatible variants, while unjustified compatibility claims can cause unsuitable purchases. Deployment should retain source evidence, permit abstention, and route consequential ambiguity to review; the present results do not validate autonomous compatibility assurance. Uneven category, language, and merchant coverage may distribute errors unevenly. No human naturalness study was performed, and no such validation is claimed. Product datasets remain subject to their publisher terms. Generative AI assisted manuscript drafting and code development; responsibility for all claims, verification, and final submission remains with the human authors.

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